A Compound-class distribution

	intermetallic	oxide	metal	chalcogenide	salt/ionic compound	halide	boride	pnictide	silicide	oxyhalide	other
MatterGen Base	48.8	14.8	0.9	1.7	2.1	4.2	2.1	3.8	2.5	4.6	14.5
MatterGen Fine-tuned	55.0	0.0	1.6	0.1	17.4	0.0	2.5	5.4	12.8	0.0	5.1
Con-CDVAE Base	14.4	21.0	1.9	14.4	1.6	14.0	0.3	4.3	0.7	13.5	13.8
Con-CDVAE Fine-tuned	18.6	0.0	4.8	1.8	14.2	0.0	27.7	9.5	6.3	0.0	16.9
CrystalFormer Base	28.9	11.7	24.5	8.8	2.4	4.2	0.7	3.3	2.0	0.4	13.1
CrystalFormer Fine-tuned	45.4	1.9	14.6	1.2	11.2	2.2	2.8	5.4	8.7	0.2	6.4
CrystaLLM Base	29.6	45.9	0.0	0.0	0.0	12.2	8.2	0.0	4.1	0.0	0.0
CrystaLLM FT	74.2	0.0	2.3	0.9	0.0	0.5	3.2	0.5	18.4	0.0	0.0
Crys-JEPA Base	30.3	16.3	1.7	13.2	4.7	11.3	2.2	4.0	5.0	3.1	8.3
Crys-JEPA Refined	76.1	0.0	20.2	0.3	0.0	0.1	0.4	0.9	0.1	0.0	1.9

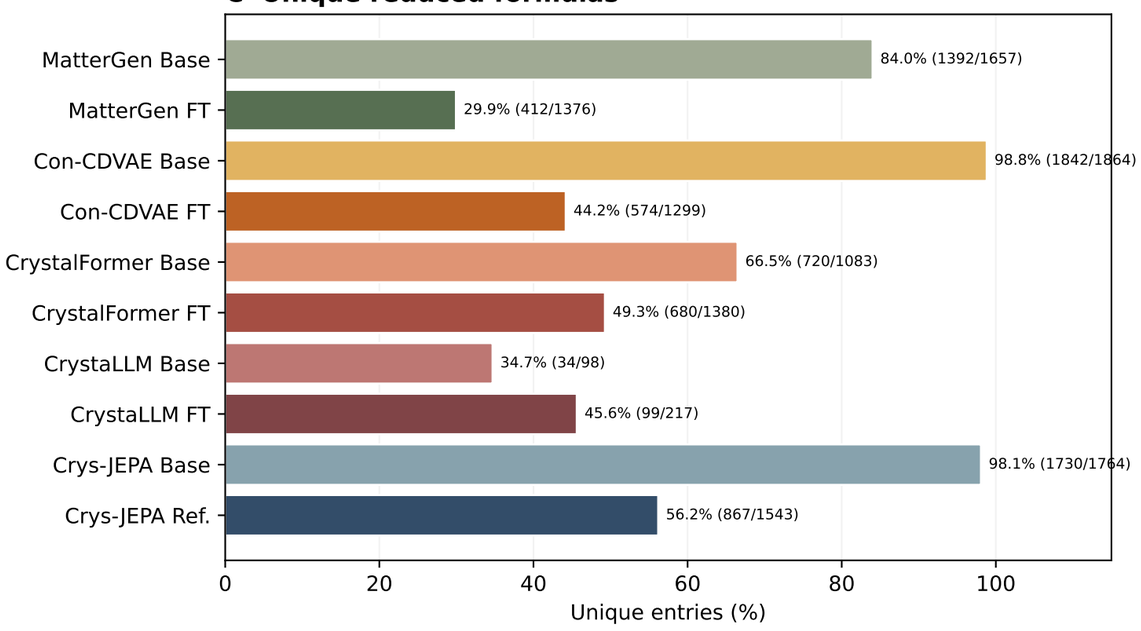
Share within cohort (%): 100, 75, 50, 25, 0

B Adaptation-associated changes

	intermetallic	oxide	metal	chalcogenide	salt/ionic compound	halide	boride	pnictide	silicide	oxyhalide	other
MatterGen	+6.2	-14.8	+0.7	-1.5	+15.3	-4.2	+0.4	+1.6	+10.3	-4.6	-9.5
Con-CDVAE	+4.2	-21.0	+2.9	-12.7	+12.6	-14.0	+27.4	+5.3	+5.6	-13.5	+3.1
CrystalFormer	+16.5	-9.8	-9.8	-7.5	+8.8	-1.9	+2.0	+2.0	+6.7	-0.2	-6.7
CrystaLLM	+44.6	-45.9	+2.3	+0.9	+0.0	-11.8	-4.9	+0.5	+14.4	+0.0	+0.0
Crys-JEPA	+45.8	-16.3	+18.5	-12.8	-4.7	-11.2	-1.8	-3.1	-4.9	-3.1	-6.4

Fine-tuned/Refined minus Base (pp): 45.92, 22.96, 0.00, −22.96, −45.92

C Unique reduced formulas



D Unique formula–space-group pairs

